COMMUNICATION WITH STRATEGIC FACT-CHECKING

Aleksandr Levkun
WU (Vienna University of Economics and Business)

November 27, 2025

Abstract

We examine communication between an informed sender and an uninformed receiver with a presence of a strategic fact-checker. The sender makes a claim about an issue to persuade the receiver to approve the sender's proposal. The fact-checker has its own goal and chooses a stochastic fact-checking policy that checks sender's claims at a cost. We characterize the fact-checker's optimal verification policy and show that it features full or partial checking depending on verification costs and on which mistakes fact-checking can actually correct. Optimal checking intensity depends on a single aspect of the environment: how skeptical the receiver is when persuasion is difficult, and how often harmful proposals arise when persuasion is easy. The receiver need not prefer a fact-checker with preferences aligned with the receiver to one with opposed preferences. With multiple fact-checkers, free-riding can reduce information transmission relative to a single fact-checker, though sufficiently high verification costs eventually reverse the comparison.

JEL Classification Numbers: C72, D82, D83

Keywords: Fact-checking, Information, Lie detection, Strategic mediation

I would like to thank Renee Bowen, Simone Galperti, and Joel Sobel for their support and guidance. I greatly appreciate the comments and suggestions from S. Nageeb Ali, Songzi Du, Florian Ederer, Shachar Kariv, Charles Louis-Sidois, Mark Machina, Denis Shishkin, Maria Titova, Joel Watson, and seminar participants at various universities. All errors are mine.

1 Introduction

Fact-checking of prominent public figures has become ubiquitous. Initially the fact-checkers focused attention on the US elections. Now they constantly check political claims over the variety of challenging topics. Undoubtedly, fact-checking has become an integral part of political discussion in the US (Graves, 2016).[1] The goal of fact-checking is to hold politicians accountable for spreading deceitful claims (Graves, 2016). However, the role of "arbiters of truth" has drawn criticism on multiple counts including the fact-checkers' bias.[2] Ostermeier (2011) points out the lacking transparency in how checked claims get selected. The selection effect may lead to a biased perception of a politician's credibility: actors who receive more negative fact-checking ratings deemed less truthful than those who are checked rarely and receive fewer negative ratings (Uscinski and Butler, 2013; Uscinski, 2015). For these reasons, our understanding of the effects of potentially biased fact-checking is important, especially in the age of fake news and alternative facts (Allcott and Gentzkow, 2017). This paper takes seriously the possibility of a strategically motivated fact-checker. We ask the following questions. Who benefits from fact-checking? How do these benefits depend on the fact-checker's preferences? Is fact-checking effective in preventing the speaker from spreading false claims? What kind of a fact-checker is preferred by a receiver and does adding fact-checkers help this receiver to learn the truth more often?

We show that the value and consequences of fact-checking depend critically on which mistakes verification can correct and what the fact-checker cares about. Fact-checking improves decision-making but can either help or harm the speaker depending on whether persuasion of uninformed receivers is hard or easy. A strategic fact-checker chooses how aggressively to verify claims based on its own objectives, and these incentives need not align with the receiver's: in some environments, the receiver prefers a "partisan" fact-checker to a neutral one. Finally, introducing additional fact-checkers improves information

[1] Graves and Cherubini (2016) document the rise of fact-checking in Europe. Major social-media platforms now incorporate fact-checking tools, most recently through third-party labels and community-driven systems, prompting concerns about how claims are selected for scrutiny and whether fact-checkers introduce bias.

[2] Examples of other critiques include an inability to fight motivated reasoning (Walter et al., 2020) and examining claims that cannot be checked reliably (Uscinski and Butler, 2013).

transmission only for claims that are very costly to verify; otherwise, free-riding reduces overall verification. Together, these results provide a new perspective on the strategic role of fact-checkers in communication.

To formalize these insights, we incorporate a strategic fact-checker in a model of cheap-talk communication. The receiver has to accept or reject a sender's proposal, but does not know whether it is good or bad for her. The sender is informed about a binary receiver's value of acceptance and can either convey this to the receiver using cheap-talk claims or stay silent. However, the sender would like the receiver to always accept and, thus, makes a claim in an attempt to persuade the receiver. The fact-checker can respond by deploying a costly verification technology. Specifically, it commits to a stochastic fact-checking policy: for each non-silent claim, it chooses a verification intensity, the probability with which a check yields a conclusive truth assessment.[3] The fact-checker maximizes its expected payoff net of the fact-checking cost, and the fact-checker's payoff function is the central object in our analysis. We consider three natural cases. First, the pro-receiver fact-checker maximizes the receiver's payoff. Second, the pro-sender fact-checker wishes for the sender's proposal to be accepted. Third, the anti-sender fact-checker wants the sender's proposal to be rejected.

Without fact-checking, pooling compromises communication: the bad sender's type pretends to be the good sender. The fact-checker is able to provide separation, thereby increasing the receiver's payoff relative to the sender-receiver cheap-talk game. However, the benefits of fact-checking for the sender depend on whether sender's information is persuasive, that is, whether the receiver accepts the sender's proposal under no communication. We show that when sender's information is not persuasive, fact-checking determines the extent to which the good sender can convince the receiver to accept. Consequently, more frequent fact-checking increases the sender's payoff. In any equilibrium, the good sender

[3]We take a stance on the capacity of the fact-checker to make strategic decisions. Graves (2017) itemizes typical steps of a process of fact-checking based on the author's field experience with three major fact-checking organizations: PolitiFact, FactCheck.org, and Washington Post's Fact Checker. The first step identifies claims to check. Then fact-checkers gather the evidence, assess the claim veracity, and publish the fact-check output transparently. We allow the fact-checker to be strategic only about the first step of this process.

simply sends the most checked claim and the players' equilibrium payoffs are unique. When sender's information is persuasive, fact-checking determines the extent to which the bad sender can dupe the receiver into accepting. As a result, more frequent fact-checking can only harm the sender. The defining property of any equilibrium is that the bad sender prefers as little fact-checking as possible but still needs to mimic the good sender's type. The good sender always gets his proposal accepted, with an opportunity to make any claim. Subsequently, various good sender's behavior corresponds to different players' payoffs.

Our first main result characterizes the optimal fact-checking policy chosen by a strategic fact-checker. Whether the fact-checker chooses to verify claims at all depends on how valuable it is to correct the receiver's mistakes in the one situation where fact-checking can actually change behavior. When the sender's information is not persuasive, so the receiver is skeptical and tends to reject, the useful role of fact-checking is to help the good sender overcome this skepticism. The fact-checker checks only if it benefits from helping the receiver accept more often, and the optimal checking frequency depends solely on how hard it is to persuade the receiver, not on how likely the good state is. By contrast, when the sender's information is persuasive, fact-checking is useful to prevent the receiver from being misled. In this case, the checking intensity depends exclusively on how frequently bad proposals arise, not on how skeptical the receiver is. In either environment, if verification is sufficiently cheap, the fact-checker checks every relevant claim. If verification is relatively costly, the fact-checker checks only occasionally. The strength of incentives to check is governed by the environment: skepticism drives fact-checking when trusted messages need verification, whereas prevalence of harmful proposals drives fact-checking when the receiver is too trusting.

The structure of the optimal fact-checking policy depends critically on the fact-checker's preferences. A fact-checker who wants the receiver to make better decisions checks only when it values correcting the receiver's mistake in the state fact-checking can influence. In contrast, the pro-sender fact-checker never checks when the receiver already accepts under no information, and the anti-sender fact-checker never checks when the receiver rejects without information. Perhaps surprisingly, a fact-checker who maximizes the receiver's payoff need not be the one the receiver prefers. When the sender's benefit from swaying the receiver's action outweighs the receiver's own benefit from learning the truth, a partisan fact-checker chooses more aggressive verification. In such cases, the receiver is better off with a biased fact-checker than with a neutral pro-receiver one, because the partisan fact-checker is more willing to incur verification costs in the state that matters for correcting the receiver's decision.

Our second main result concerns whether adding more fact-checkers improves the quality of communication. The answer is negative unless verification is sufficiently costly. When two fact-checkers choose verification policies simultaneously, each prefers the other to bear the cost of producing informative checks. This generates a free-riding motive: a strategic fact-checker wants to benefit from more informative communication without paying for it. If verification is cheap, all equilibria are asymmetric: one fact-checker performs all checking while the other remains inactive, providing no gain relative to having a single fact-checker. When verification is moderately costly, neither fact-checker is willing to check as aggressively as it would alone. Instead, both choose strictly positive but reduced checking intensities in a unique interior equilibrium, and the resulting verification probability is strictly below that of a single dedicated fact-checker. Thus, multiple fact-checkers do not necessarily strengthen accountability. Instead, they create a coordination problem that leads to an underprovision of verification. Only when verification becomes sufficiently costly, free-riding incentives dampen as each fact-checker believes that the other is increasingly likely to fail to check. When facts are sufficiently difficult to verify in this sense, information transmission improves with multiple fact-checkers.

Several authors suggested that partisan fact-checkers can be harmful for more informative political discourse (Ostermeier, 2011; Graves, 2016). We show that this is not necessarily the case. The partisan fact-checker may be willing to fact-check the claim to help or hurt the sender, while the fact-checking cost may prevent the non-partisan fact-checker from selecting this claim. As for the fact-checking cost, automation of fact-checking will necessarily drive it down. While most fact-checking efforts are currently made by journalists and experts, there is hope for systematic computer-assisted fact-checking (Hassan et al., 2017; Graves, 2018). Our results suggest that decreasing the verification cost sustains more informative communication. However, currently researchers and practitioners agree that the real promise of the automated fact-checking lies in methods to assist human fact-checkers in selecting the claims for verification (Graves, 2018). This may "debias" the fact-checker, which can have adverse effects for information transmission.

Related Literature. This paper contributes to the growing literature on communication with detectable deception. Several recent papers study cheap-talk games in which false statements may be exposed exogenously. In Balbuzanov (2019) and Dziuda and Salas (2018), the receiver observes a signal that reveals lies with a fixed probability and higher detection makes communication more informative. In Balbuzanov (2019), small detection probabilities can sustain fully revealing equilibria. In Dziuda and Salas (2018), the moderate and high sender types tell the truth, while low types lie claiming to be the high types. A similar equilibrium structure is obtained in Sadakane and Tam (2022) that allow the receiver to pay to inspect claims. Holm (2010) shows that lie- or truth-detection shrinks the set of equilibria in bluffing games and eliminates multiplicity then detection is sufficiently likely. Relative to this work, our fact-checking technology can verify both truths and lies, and crucially the detection probabilities are not exogenous. Instead, they are chosen strategically by a fact-checker who bears verification costs and whose preferences shape the informativeness of communication. Our focus is thus not on the consequences of better lie detection, but on how a strategic intermediary chooses and deploys verification.[4]

This paper also connects to the literature on optimal auditing pioneered by Townsend (1979). As in Border and Sobel (1987) and Mookherjee and Png (1989), our fact-checker commits to inspection probabilities, and as in Baron and Besanko (1984) and Laffont and Tirole (1986), verification may be noisy and fail to uncover the true state. A key distinction, however, is that auditors use transfers or penalties to discipline misreporting, whereas our fact-checker has only informational tools: by choosing verification intensity, it reshapes the sender-receiver interaction rather than enforcing truth-telling directly. In this light, our framework bridges the optimal auditing literature and the literature on communication

[4] Andreottola and Louis-Sidois (2025) analyze a related model where politicians have asymmetric access to facts and biased fact-checkers scrutinize one side more heavily. Their strategic selective scrutiny differs from our choice of verification intensity but reinforces the idea that fact-checker motives shape claim selection.

with detectable deceit.

Our model also relates to the literature on strategic mediation. In Ivanov (2010), a strategic mediator with commitment power can garble messages to implement the receiver-optimal outcome, while Salamanca (2021) studies a mediator who maximizes the sender's payoff. Our fact-checker also acts as a strategic intermediary with commitment power but faces a constraint: rather than being able to design arbitrary message structures, it can only influence communication via a costly verification technology tied to the sender's claims. This makes the problem close in spirit to Bayesian persuasion (Kamenica and Gentzkow, 2011), and our framework studies persuasion under an informational production constraint. Relatedly, Ederer and Min (2024) incorporate lie detection in Bayesian persuasion and show that improved detection may hurt the sender. In our setting, more aggressive verification benefits the sender by facilitating separation when persuasion is hard.

Finally, our paper connects to the empirical literature on fact-checking, recently surveyed by Nieminen and Rapeli (2019). Evidence on the effectiveness of fact-checking is mixed. While several studies show that corrections improve belief accuracy (Weeks and Garrett, 2014; Weeks, 2015; Barrera et al., 2020), others document limited impact on attitudes or persistent heterogeneity driven by ideology.[5] Nyhan and Reifler (2015) demonstrate that the fact-checking efforts may discourage politicians from spreading false claims. Regarding who conducts fact-checking, Wintersieck, Fridkin, and Kenney (2021) find small effects of the source on assessments of accuracy, while Lim (2018) documents that different fact-checkers rarely scrutinize the same claims.[6] Louis-Sidois (2025) provides evidence that fact-checkers' claim selection and scrutiny intensity correlate with the ideological leanings of affiliated outlets. Our results complement this empirical evidence by showing that fact-checker's objectives systematically shape verification intensity, and that when multiple fact-checkers operate, free riding can lead to strictly lower overall verification than a single dedicated fact-checker provides.

[5] Nyhan and Reifler (2010) identify a "backfire effect" among some ideological groups, though subsequent work finds little evidence for its robustness (Weeks and Garrett, 2014; Nyhan, Porter, et al., 2020).

[6] See Amazeen (2015) and Amazeen (2016) for evidence of fact-check output consistency, and Marietta, Barker, and Bowser (2015) for variation on topics of climate change, racism, and national debt.

2 Model

There are three players, a sender (he), a fact-checker (it), and a receiver (she), who participate in a one-round communication game. The receiver chooses whether to accept or reject the sender's proposal. Her payoff depends on the state of the world, whereas the sender has state-independent preferences for approval. The receiver's decision relies on information contained in a sender's claim and a fact-check output. A fact-checking policy specifies, for each claim, how the fact-checker will investigate it. Successful fact-checking reveals whether the sender's claim is truthful, while unsuccessful fact-checking generates the empty output. The fact-checker can commit to its policy in advance and chooses it to maximize its own payoff, taking into account the costly process of successful verification.

Players and information. There is an issue $\theta \in \{0,1\}$ that is relevant for a receiver's decision between accepting ($a = A$) or rejecting ($a = R$) the sender's proposal. Nature draws θ from the prior distribution with probability $\mu(\theta)$, where $\mu(1) = \mu \in (0,1)$. The privately informed θ-sender learns θ and sends a costless message $m \in \mathcal{M} = \{0,1,m_s\}$, where $m = m_s$ is a *silent message*, and $m \in \{0,1\}$ is a claim that $\theta = m$. The fact-checker decides whether to check m for veracity using a fact-checking technology described below. Successful fact-checking generates the fact-check output $\mathcal{O} = \mathcal{T}$ if m is truthful and $\mathcal{O} = \mathcal{F}$ if m is deceitful. Unsuccessful fact-checking generates an empty output, $\mathcal{O} = \varnothing$. The receiver observes message m and fact-check output $\mathcal{O}$ and then acts, $a \in \{A,R\}$.

Fact-checking technology. The fact-checker has access to a technology that verifies the truthfulness of sender's claims. For each non-silent message $m \in \{0,1\}$, the fact-checker selects a probability $p(m) \in [0,1]$, with which the investigation produces a conclusive output. When a conclusive check is generated, the output is $\mathcal{O} = \mathcal{T}$ when $\theta = m$ and $\mathcal{O} = \mathcal{F}$ when $\theta \neq m$. With the remaining probability $1 - p(m)$, verification fails and $\mathcal{O} = \varnothing$. The choice of the probability of a conclusive fact-check is costly: choosing $p(m)$ entails a verification cost given by an increasing and convex cost function $C(p(m))$, with $C(0) = 0$. We interpret this as the realized operational cost of conducting a more thorough investigation, incurred only when a message is checked. For tractability, we specify $C(p) = \frac{cp^2}{2}$, where $c > 0$ measures how costly fact-checking effort is at the margin; but our qualitative results remain unchanged for more general $C(\cdot)$. If the message is silent, $m = m_s$, then no costly fact-checking takes place and the output is always $\mathcal{O} = \varnothing$.

Strategies. A sender's strategy is a probability distribution $\sigma(\cdot|\theta)$ over messages $m \in \mathcal{M}$. The fact-checker's strategy is a *fact-checking policy* given by a function $p : \{0,1\} \to [0,1]$, where $p(m)$ specifies the probability of a conclusive fact-check of non-silent m. Without loss, we can set $p(m_s) = 0$. Finally, a receiver's acceptance strategy $\alpha(m,\mathcal{O})$ specifies the probability of choosing $a = A$ after observing a message m and a fact-check output $\mathcal{O}$. The receiver's posterior belief that $\theta = 1$ following $(m,\mathcal{O})$ is denoted as $\pi(m,\mathcal{O})$.

Payoffs. The sender's objective is to convince the receiver to accept: his payoff is $u_S(a) = \mathbf{1}\{a = A\}$. The receiver's payoff $u_R(a,\theta)$ is $\theta - \omega$ if she accepts and 0 if she rejects the sender's proposal.[7] The parameter $\omega \in (0,1)$ is the minimal belief that $\theta = 1$ for the receiver to be willing to accept the sender's proposal. The fact-checker has preferences over action-state pairs, $u_F(a,\theta)$, net of the checking cost. We will consider three natural variations of fact-checker's preferences: the fact-checker is *pro-receiver* if $u_F(a,\theta) = u_R(a,\theta)$, *pro-sender* if $u_F(a,\theta) = u_S(a)$, and *anti-sender* if $u_F(a,\theta) = -u_S(a)$. The fact-checker's preferences are fixed, parameters ω, μ, and the cost function $C(\cdot)$ are common knowledge. All players are expected utility maximizers.

Solution concept and equilibrium. We assume that the fact-checker has commitment power. Accordingly, the fact-checker chooses the fact-checking policy p at the outset of the game.[8] Each fact-checker's choice of fact-checking policy p initiates a subgame between

[7] The same specification appears in Kolotilin et al. (2017), Shishkin (2025), among others. It makes $a = A$ a "risky" action with a state-dependent payoff, while $a = R$ is a "safe" action.

[8] Note that in this setting, the fact-checking policy is only relevant for the sender's strategy, whereas the receiver may potentially not even observe p. The situation will change if the receiver has an option to search for a fact-check at some search cost. Then the decision whether to search would take p into account.

the sender and the receiver for which we impose standard perfect Bayesian equilibrium conditions together with *consistency with fact-checking technology*:

1. If at least one of $\sigma(m|0)$ or $\sigma(m|1)$ is non-zero, then $\pi(m,\varnothing) = \frac{\mu\sigma(m|1)}{\mu\sigma(m|1)+(1-\mu)\sigma(m|0)}$.
2. For any non-silent $m \in \{0,1\}$, $\pi(m,\mathcal{T}) = \mathbf{1}\{m=1\}$ and $\pi(m,\mathcal{F}) = \mathbf{1}\{m=0\}$.
3. If $\pi(m,\mathcal{O}) > \omega$, then $\alpha(m,\mathcal{O}) = 1$. If $\pi(m,\mathcal{O}) < \omega$, then $\alpha(m,\mathcal{O}) = 0$.
4. $\sigma(\cdot|\theta)$ is supported on

$$\operatorname*{argmax}_{m\in\mathcal{M}} \{p(m)\alpha(m,\mathcal{T})\mathbf{1}\{\theta=m\} + p(m)\alpha(m,\mathcal{F})\mathbf{1}\{\theta\neq m\} + (1-p(m))\alpha(m,\varnothing)\}\ .^{9}$$

The first requirement is Bayesian updating of the receiver's beliefs after observing on-path messages. Consistency with fact-checking technology requires that the receiver correctly interprets any nonempty fact-check output for both on-path and off-path messages.[10] The third requirement states that the receiver's decision is optimal given her beliefs. The final requirement prescribes that each type of sender only uses messages that maximize his probability of acceptance, with the understanding that these messages may be checked.

Given a fact-checking policy p, we refer to a triple (σ,α,π) that satisfies above conditions as a *p-equilibrium*. Let $\mathcal{E}(p)$ denote the set of p-equilibria, with a typical element ϵ. Each p-equilibrium ϵ induces a joint distribution of actions and states $\lambda(a,\theta|\epsilon,p)$.[11] The fact-checker's problem is to choose a fact-checking policy p and p-equilibrium jointly to maximize its expected payoff net of the fact-checking cost:

$$\max_{p} \max_{\epsilon\in\mathcal{E}(p)} \left\{ \sum_{a,\theta} u_F(a,\theta)\lambda(a,\theta|\epsilon,p) - \sum_{\theta,m} \mu(\theta)\sigma(m|\theta)C(p(m)) \right\}.$$

[9] Given that $p(m_s)=0$, the value assigned to $\mathbf{1}\{\theta=m_s\}$ is irrelevant.

[10] Our definitions of on-path and off-path messages are standard: fixing σ, the on-path messages satisfy $\sigma(m|1)>0$ or $\sigma(m|0)>0$. The off-path messages are messages that are not on-path.

[11] Formally, $\epsilon=(\sigma,\alpha,\pi)$ generates a joint action-state distribution as follows:

$$\lambda(a=A,\theta|\epsilon,p) = \mu(\theta)\sum_{m\in\mathcal{M}} \sigma(m|\theta)\left[p(m)\alpha(m,\mathcal{T})\mathbf{1}\{\theta=m\} + p(m)\alpha(m,\mathcal{F})\mathbf{1}\{\theta\neq m\} + (1-p(m))\alpha(m,\varnothing)\right],$$

and $\lambda(a=R,\theta|\epsilon,p) = \mu(\theta) - \lambda(a=A,\theta|\epsilon,p)$.

A solution to this problem, p^* and $\epsilon^* \in \mathcal{E}(p^*)$, is an *equilibrium*. In our definition of equilibrium, we view the fact-checker as a principal who is able to select among its favorite equilibria.[12]

Feasible payoffs and environments. Fixing a fact-checking policy p and a p-equilibrium, let $U_S(\theta)$ denote the payoff of θ-sender and $U_S = \mu U_S(1) + (1-\mu)U_S(0)$ denote the sender's ex ante payoff. The receiver's ex ante payoff is denoted by U_R. We say that equilibrium payoffs $U_S(\theta)$, U_S, and U_R are *feasible* if there is a fact-checking policy p and a p-equilibrium that generate those payoffs.

We refer to a pair (μ,ω) as an *environment*. It will be useful to distinguish whether the environment is predisposed toward the sender. When $\mu < \omega$, that is, under no information the receiver chooses to reject the sender's proposal, we call (μ,ω) a *sender-unfavorable environment* (SUE). When the receiver accepts under the prior, $\mu > \omega$, we call (μ,ω) a *sender-favorable environment* (SFE). The knife-edge case $\mu = \omega$ behaves as a limit of either SUE or SFE depending on the equilibrium selected by the fact-checker.

3 Feasible payoffs and subgame Equilibria

We begin by characterizing the equilibrium payoffs that can arise under all possible fact-checking policies p. We describe how the sender's incentives depend on the environment and on the fact-checking policy. To build intuition, we consider two extreme cases of fact-checking policies: *no fact-checking* and *full fact-checking*.

Under the no fact-checking policy, $p(1) = p(0) = 0$, every message produces the inconclusive outcome $\mathcal{O} = \varnothing$. Messages do not carry an intrinsic meaning and the game collapses to a cheap-talk game with a binary state and state-independent sender preferences. Pooling is not deterred. In SUE, the receiver's prior belief is too low to accept under no information. Any on-path p-equilibrium message leads to the posterior belief $\pi(m,\mathcal{O}) < \omega$, so that the

[12] This is a standard assumption in the information design literature for an agent with commitment power (e.g., Kamenica and Gentzkow, 2011). Mathevet, Perego, and Taneva (2020) discuss alternative selection rules such as worst-equilibrium selection.

receiver rejects. Consequently, sender's feasible payoffs are $U_S(1) = U_S(0) = 0$. In SFE, the receiver accepts the sender's proposal after observing any on-path message m, with $\pi(m,\mathcal{O}) \geq \omega$. Consequently, sender's feasible payoffs are $U_S(1) = U_S(0) = 1$.[13]

Under the full fact-checking policy, $p(1) = p(0) = 1$, any non-silent message is successfully verified and produces either $\mathcal{T}$ or $\mathcal{F}$ as the output, with the receiver inferring the state. The silent message m_s remains unchecked. In SUE, silence prompts the receiver to reject. Thus, the 1-sender cannot benefit from staying silent and instead may send $m = 1$. This message is verified as truthful and accepted. As a result, any message sent by the 0-sender leads to rejection. Full fact-checking thus separates the sender's types, with resulting payoffs $U_S(1) = 1$ and $U_S(0) = 0$. In SFE, the receiver accepts without information, and a class of pooling-on-silence p-equilibria is sustainable. In these p-equilibria, the 0-sender always stays silent, while the 1-sender may mix between m_s and non-silent messages, and the receiver always accepts. A separating pattern is also sustainable, in which the 1-sender may send $m = 1$ and is accepted after verification, while any message of the 0-sender results in rejection. Thus, under full fact-checking in SFE, the 1-sender always gets payoff 1, while the 0-sender's payoff is either 0 or 1 depending on the p-equilibrium selection: $U_S(1) = 1$, $U_S(0) \in \{0,1\}$.

An important caveat arises under the full fact-checking technology. This case features the following p-equilibrium for any environment: the 1-sender sends a non-silent message, e.g., $m = 1$, is verified truthful, and is accepted, while the 0-sender remains silent and is rejected. This separation requires verification only of the 1-sender's message and therefore can be sustained at a lower effective fact-checking cost. Importantly, such p-equilibrium does not exist under imperfect fact-checking, since silence cannot credibly distinguish types when verification may fail. As a result, full fact-checking introduces a discontinuity in feasible outcomes and will be treated separately later.

For any fact-checking policy p, we summarize an aggressiveness of verification by two statistics. The *maximum verification intensity* $\overline{p} = \max\{p(0), p(1)\}$, is the highest probability with which any non-silent message can be conclusively checked. A correspond-

[13] Irrespective of the environment, some p-equilibria can still be informative. However, the receiver's payoff is fixed across all p-equilibria at $U_R = \max\{0, \mu - \omega\}$, since her optimal action does not change.

ing most-checked message is denoted by $\overline{m}$. Similarly, the *minimum verification intensity* $\underline{p} = \min\{p(0), p(1)\}$, is the lowest probability of conclusive verification among non-silent messages, and $\underline{m}$ denotes the least-checked non-silent message.[14]

We now characterize the feasible sender's payoffs that arise under fact-checking policies. We show that these payoffs can be summarized by the maximum verification intensity $\overline{p}$. Two insights from extreme fact-checking policies generalize to all verification intensities. First, the 0-sender's proposal is always rejected by the receiver in SUE. Second, the 1-sender always gets his proposal accepted in SFE.

Proposition 1. *Fix the maximum verification intensity* $\overline{p} = \max\{p(0), p(1)\}$*.*

- *In a sender-unfavorable environment, $\mu < \omega$, the sender's payoffs are uniquely determined: $U_S(0) = 0$ and $U_S(1) = \overline{p}$.*
- *In a sender-favorable environment, $\mu > \omega$, the 1-sender's payoff is uniquely determined, while the 0-sender's payoff varies: $U_S(1) = 1$ and $U_S(0) \in [1 - \overline{p}, 1]$. Moreover, every payoff in the admissible range is attained by some fact-checking policy p with $\overline{p}$.*

A conclusive fact-check fully reveals the state and leaves no room for receiver persuasion. Thus, the players' incentives are shaped by the likelihood that a fact-check produces a conclusive output, captured by $\overline{p}$.[15] Importantly, since the sender faces no direct penalty from being caught in a lie, the most-checked message $\overline{m}$ can be either $m = 1$ or $m = 0$. In what follows, we adopt the natural interpretation in which $\overline{m} = 1$, so that verification success corresponds to recognizing truthful claims; however, all formal results apply regardless of which message is most intensively checked.

In SUE, the 0-sender can never be accepted in equilibrium. Acceptance after an inconclusive outcome would give the 0-sender a profitable deviation and therefore cannot be sustained. Consequently, the 0-sender's payoff is uniquely determined at zero. The

[14]If $p(1) = p(0)$, either non-silent message may be designated as $\overline{m}$ or $\underline{m}$.

[15]Proposition 1 implies that the receiver always plays a pure strategy after on-path messages in both SUE and SFE. Indeed, if the receiver were mixing on the equilibrium path, the payoffs of both sender's types would be interior, which contradicts Proposition 1.

1-sender is accepted only when verification succeeds. Higher verification intensities make it more likely that honesty is recognized, strictly benefiting the 1-sender. The extremes coincide with our earlier benchmarks: $\overline{p} = 0$ leads to no gain from communication, while $\overline{p} = 1$ gives full credibility to the 1-sender.

In SFE, the 1-sender can always secure acceptance. Fact-checking plays a different role here: it disciplines the 0-sender. Whenever a non-silent message is conclusively checked, the 0-sender is revealed and rejected. Thus, his payoff reflects the probability that verification fails. As $\overline{p}$ increases, deception becomes more likely to be exposed, and the 0-sender's payoff decreases. Formally, $U_S(0) = 0$ ranges from 1 (when he is never checked) to $1 - \overline{p}$ (when he always uses the most-checked message). Unlike in SUE, this payoff is not unique: different fact-checking policies and p-equilibria sustain different levels of exposure risk.

In SUE, fact-checking only benefits the 1-sender by making truthful persuasion more effective; in SFE, fact-checking only harms the 0-sender by making mimicking and deceptive persuasion less effective. The single parameter $\overline{p}$ summarizes these forces and determines the sender's feasible welfare under any fact-checking policy.

We now relate our result to the best possible communication outcome for the sender. In a setting without fact-checking but with the sender's commitment power as in Kamenica and Gentzkow (2011), the sender can obtain the ex ante payoff of $\frac{\mu}{\omega}$ in SUE. This outcome relies on a fine-tuned randomization by the 0-sender that keeps the receiver exactly indifferent after the "winning" message. In our setting, the best payoff the sender can achieve in SUE is μ under full fact-checking. The commitment payoff is unattainable because randomization by the 0-sender is not credible: any attempt to mix across messages is potentially exposed by verification, and the sender has no commitment power to enforce such mixing. Thus, fact-checking limits persuasion relative to the commitment benchmark. By contrast, in SFE, the comparison is trivial: since the receiver already prefers acceptance absent information, the sender can replicate the commitment outcome by staying silent. Importantly, with a richer state space, optimal persuasion may no longer require sender-side randomization, and fact-checking may enable commitment.[16] We discuss this in more detail in Section 6.

[16] Titova and Zhang (2025) show that the sender can achieve the commitment outcome with verifiable information only. Also related is Guo and Shmaya (2021) where the sender incurs "miscalibration cost" for

Proposition 1 shows that fact-checking affects the sender's welfare by varying only one type's payoff in each environment. First, the 0-sender is not able to escape the zero payoff in SUE regardless of how often messages are verified. Additional fact-checking can only help the 1-sender to gain credibility. Second, the 1-sender is always capable of getting his proposal accepted in SFE irrespective of the 0-sender's strategy and the fact-checking policy. Additional fact-checking can only expose the 0-sender more frequently. We now ask a natural question: when the fact-checker checks more aggressively, how are the ex ante payoffs of the sender and the receiver affected? We say that a fact-checking policy p is *more aggressive* than p' if $\overline{p} > \overline{p}'$, that is, the maximum verification intensity under p exceeds that under p'.[17] The next proposition shows how increasing aggressiveness alters sender and receiver welfare.

Proposition 2. *For the fixed maximum verification intensity $\overline{p}$, the ex ante payoffs are: $U_S = \mu\overline{p}$ and $U_R = \mu(1-\omega)\overline{p}$ in a sender-unfavorable environment (SUE); $U_S \in [1-(1-\mu)\overline{p}, 1]$ and $U_R \in [\mu-\omega, \mu-\omega+(1-\mu)\omega\overline{p}]$ in a sender-favorable environment (SFE). Hence, when the fact-checking policy is more aggressive in the sense of increasing $\overline{p}$:*

- *both the sender and the receiver benefit in SUE,*
- *the lower bound on the sender's payoff decreases and the upper bound on the receiver's payoff increases in SFE.*

Players' incentives rely not on the entire specification of fact-checking but on how often the most-checked message can generate conclusive evidence. In SUE, persuasion is intrinsically difficult. Fact-checking helps only the 1-sender, while the 0-sender can never be accepted, even after inconclusive checks. Thus, his payoff is pinned at zero. By contrast, the 1-sender succeeds only when verification is complete. The receiver benefits in exactly the same event. Thus, the ex ante payoffs are $U_S = \mu\overline{p}$ and $U_R = \mu(1-\omega)\overline{p}$. Hence, more aggressive fact-checking benefits both players in SUE by helping the 1-sender separate more often.

undermining the meaning of a claim. High miscalibration cost acts as a substitute for commitment.

[17]This order is adopted for expository purposes. An alternative definition would require $p(0) \geq p'(0)$ and $p(1) \geq p'(1)$, with at least one strict inequality. Our conclusions would remain unchanged.

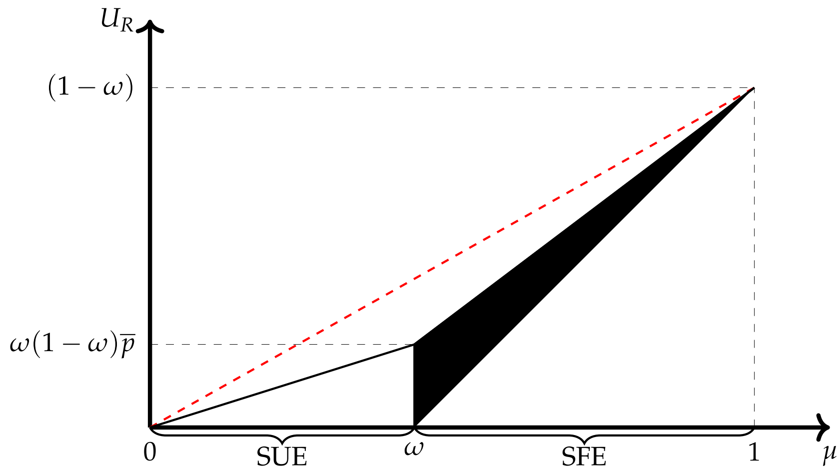

Figure 1: Feasible U_R depending on prior μ for fact-checking policies with fixed $\overline{p}$. The dashed red line corresponds to the receiver's payoff under complete information.

In SFE, acceptance is already optimal under the prior. The 1-sender always secures acceptance and is unaffected by $\overline{p}$. Fact-checking instead disciplines the 0-sender: when verification is conclusive, deception is exposed and hence leads to rejection. Thus, the 0-sender's payoff depends on how often he can hide behind inconclusive verification. When $\overline{p}$ increases, exposure becomes more likely and his worst-case payoff goes down. The receiver gains for the same reason: fact-checking prevents regrettable acceptance when $\theta = 0$. Furthermore, equilibrium multiplicity implies that payoffs in SFE are not unique. Pooling on silence generates the highest sender's payoff and the lowest receiver's payoff. Pooling on the most-checked message generates the lowest sender's payoff and the highest receiver's payoff. Pooling on $\underline{m}$ generates interior payoffs. In all cases, more aggressive fact-checking shrinks the sender's feasible payoff set and expands the receiver's.

Figure 1 provides an illustration of the part of Proposition 2 on the receiver's payoff. The receiver is better off when the fact-checking policy is more aggressive as it sustains more informative communication. The receiver's payoff under complete information is attainable only when $\overline{p}$ approaches one, which can be achieved by the full fact-checking policy. Across both environments, the maximum verification intensity captures how effectively verification disrupts pooling. In SUE, pooling harms the 1-sender, so higher $\overline{p}$ enables credible and beneficial separation. In SFE, pooling protects the 0-sender, so higher $\overline{p}$ undermines deception and improves receiver's welfare. Thus, increasing aggressiveness improves informativeness, but its distributional consequences reverse across environments.

4 Optimal fact-checking

In this section, we characterize the optimal fact-checking policy for a fact-checker with arbitrary preferences over receiver's action and states. This allows us to compare different fact-checkers from the receiver's perspective.

The fact-checker chooses a fact-checking policy p and can select among p-equilibria. Its objective is to maximize expected payoff net of verification costs:

$$\max_{p} \max_{\epsilon \in \mathcal{E}(p)} \left\{ \sum_{a,\theta} u_F(a,\theta)\lambda(a,\theta|\epsilon,p) - \sum_{\theta,m} \mu(\theta)\sigma(m|\theta)C(p(m)) \right\},$$

where $\lambda(a,\theta|\epsilon,p)$ in the induced distribution of actions and states and $C(p) = \frac{cp^2}{2}$ is the cost of choosing verification intensity p for message m.

Proposition 1 implies that $\lambda(a,\theta|\epsilon,p)$ depends on p only through the maximum verification intensity $\overline{p}$. As a result, without loss of generality, we can treat the fact-checker as choosing $\overline{p} \in [0,1]$ and, conditional on $\overline{p}$, selecting its favorite p-equilibrium, which jointly determines the induced outcome distribution and the minimal expected cost of implementing $\overline{p}$. Specifically, we say that the fact-checker *implements* $\overline{p}$ if it induces p-equilibrium under some policy p with $\overline{p}$ such that, when $\overline{p} < 1$, message $\overline{m}$ is always used in relevant pooling situations: in SUE, the 1-sender always sends $\overline{m}$; in SFE, both sender types always send $\overline{m}$. Thus, implementing $\overline{p} < 1$ means that a conclusive fact-check occurs with probability $\overline{p}$ precisely in the situations where the sender attempts to pool, ensuring that verification has real disciplinary bite rather than being irrelevant. Let $K(\overline{p})$ denote the minimal cost of implementing $\overline{p}$. The next Lemma characterizes $K(\overline{p})$.

Lemma 1. *The minimal cost of implementing the maximum verification intensity $\overline{p}$ is*

- $K(\overline{p}) = \frac{\mu}{\omega}C(\overline{p})$ *for* $\overline{p} < 1$ *and* $K(1) = \mu C(1)$ *in a sender-unfavorable environment;*
- $K(\overline{p}) = C(\overline{p})$ *for* $\overline{p} < 1$ *and* $K(1) = \mu C(1)$ *in a sender-favorable environment.*

Lemma 1 highlights a simple insight: to make imperfect fact-checking meaningful, the fact-checker must induce pooling on the most-checked message. The cheapest p-equilibrium is the one in which the sender types place the minimal required probability on $\overline{m}$ while sustaining players' incentives. In SUE, only the 1-sender needs to use $\overline{m}$ for fact-checking to matter, and the 0-sender's presence on $\overline{m}$ can be kept small enough while justifying rejection after inconclusive checks. In SFE, however, both types must fully pool on $\overline{m}$ for verification to deter deception, so that checks occur with probability $\overline{p}$. At $\overline{p} = 1$, a cheap separating equilibrium becomes available in both environments: only the 1-sender is ever checked, while the 0-sender stays silent, reducing the expected frequency of costly verification to μ. Thus, Lemma 1 shows that cost-effective implementations are the ones exerting just enough discipline to make fact-checking bite, while minimizing how often costly checks occur.

Equipped with Lemma 1, we are ready to characterize the optimal fact-checking policy for a fact-checker with general preferences $u_F(a,\theta)$. It is convenient to define define $\Delta_+ = u_F(A,1) - u_R(A,1)$ and $\Delta_- = u_F(R,0) - u_R(A,0)$, the fact-checker's gain from inducing acceptance when the state is good and from inducing rejection when the state is bad, respectively. Indeed, by our analysis of p-equilibria, fact-checking affects outcomes only in a single state, $\theta = 1$ in SUE and $\theta = 0$ in SFE. As a result, some fact-checking can be optimal only if it is beneficial for the fact-checker: $\Delta_+ > 0$ in SUE and $\Delta_- > 0$ in SFE. In this case, the optimal fact-checking trades-off its benefit of more accurate receiver's actions against the effective verification cost. We show that full fact-checking is optimal when verification is cheap, while interior verification intensities arise when verification is sufficiently costly.

Proposition 3. *Let $\Delta_+ = u_F(A,1) - u_F(R,1)$ and $\Delta_- = u_F(R,0) - u_F(A,0)$ denote the fact-checker's gain from inducing acceptance when $\theta = 1$ and from inducing rejection when $\theta = 0$, respectively. The optimal fact-checking policy is given by the optimal maximum verification intensity $\overline{p}$, characterized as follows.*

- *In SUE, if $\Delta_+ \leq 0$ then no fact-checking is optimal, $\overline{p} = 0$. If $\Delta_+ > 0$, define the threshold*

$c_{SUE}(u_F) = \Delta_+(1+\sqrt{1-\omega})$. *Then the optimal intensity is*

$$\overline{p} = \begin{cases} 1\,, & \text{if } 0 < c \leq c_{SUE}(u_F)\,, \\ \dfrac{\omega\Delta_+}{c}\,, & \text{if } c > c_{SUE}(u_F)\,. \end{cases}$$

- *In SFE, if* $\Delta_- \leq 0$ *then no fact-checking is optimal,* $\overline{p} = 0$*. If* $\Delta_- > 0$*, define the threshold* $c_{SFE}(u_F) = \frac{(1-\mu)\Delta_-}{\mu}(1+\sqrt{1-\mu})$*. Then the optimal intensity is*

$$\overline{p} = \begin{cases} 1\,, & \text{if } 0 < c \leq c_{SFE}(u_F)\,, \\ \dfrac{(1-\mu)\Delta_-}{c}\,, & \text{if } c > c_{SFE}(u_F)\,. \end{cases}$$

This result has several important implications. First, the optimal fact-checking policy takes an intuitive form. In either environment, the fact-checker compares the benefit of correcting the receiver's action in the one state where fact-checking can alter behavior against the effective verification cost. In particular, when the fact-checker attaches no value to correcting receiver's actions ($\Delta_+ \leq 0$ in SUE and $\Delta_- \leq 0$ in SFE), no fact-checking is optimal. When the fact-checker values improved decision-making, the optimal verification intensity is either full or interior, depending on whether the cost of verification is low or high, respectively. Thus, strategic fact-checking is deployed only when it helps correct the receiver's decision of rejecting too often in SUE and accepting too often in SFE to the extent justified by verification cost.

Second, a notable feature of the optimal checking intensity is that it depends on only one primitive of the environment. In SUE, fact-checking matters only for $\theta = 1$, and the separation it provides is relevant because the receiver rejects too often under no communication. Thus, the optimal policy depends on how hard it is to sustain persuasion in $\theta = 1$, fully captured by the receiver's acceptance threshold ω. Fact-checking becomes less valuable the harder it is to persuade the receiver, so full verification is optimal only if verification is sufficiently cheap. However, when the receiver is harder to persuade, conditional on optimal imperfect checking, the fact-checker increases the intensity to overturn the receiver's pessimism. In SFE, fact-checking matters only for $\theta = 0$, and the detection it provides is relevant because of the excessive acceptance. As a result, the optimal policy depends on the prevalence of $\theta = 0$ but not on ω, since the receiver accepts under no information. When μ increases, preventing regretful acceptance in $\theta = 0$ becomes less valuable. Thus, full fact-checking is optimal only at lower cost levels and conditional on high cost the optimal interior intensity decreases.

Third, the optimal fact-checking policy features discontinuity at a cost threshold. As the cost parameter approaches the threshold from above, the optimal fact-checking intensity is interior, but at a threshold it jumps to full fact-checking. This jump is driven by the availability of a low-cost separating equilibrium at $\overline{p} = 1$: once perfect verification is chosen by the fact-checker, the sender types can separate with only the 1-sender using the checked message and the 0-sender staying silent. Hence, the availability of cheap separation makes full verification more attractive than a sufficiently high but imperfect intensity.

Finally, Proposition 3 allows us to describe receiver's preferences over settings with different fact-checker's preferences. To fix ideas, suppose that the fact-checker's payoff is a linear combination of sender and receiver payoffs: $u_F(a,\theta) = \beta_S u_S(a) + \beta_R u_R(a,\theta)$, where β_S and β_R parameterize how sympathetic the fact-checker is to the sender or to the receiver. This specification allows us to determine the receiver's preferences over fact-checkers in terms of weights β_S and β_R, as the following corollary shows.

Corollary 1. *Suppose $u_F(a,\theta) = \beta_S u_S(a) + \beta_R u_R(a,\theta)$. The receiver benefits when:*

- *β_S increases and β_R increases in a sender-unfavorable environment,*
- *β_S decreases and β_R increases in a sender-favorable environment.*

Corollary 1 shows that the receiver prefers fact-checkers whose preferences induce more aggressive verification in the states where she is prone to error. In SUE, she rejects too often, and fact-checking only helps when $\theta = 1$. Thus, a fact-checker who values the sender's acceptance payoff or the correct receiver's acceptance more benefits the receiver. In SFE, she accepts too often, and fact-checking only matters when $\theta = 0$. Accordingly, a fact-checker who cares less about the sender and more about the correct receiver's rejection checks more aggressively.

We can specialize even more and consider the receiver's preferences over pro-receiver, pro-sender, and anti-sender fact-checkers. The pro-receiver fact-checker puts a weight of $\beta_S = 0$ on the sender payoff and a weight of $\beta_R = 1$ on the receiver payoff. The pro-sender fact-checker's weights are $(\beta_S, \beta_R) = (1, 0)$ and the anti-sender fact-checker's weights are $(\beta_S, \beta_R) = (-1, 0)$. The following corollary shows that the receiver-preferred fact-checker is pro-sender in SUE and anti-sender in SFE.

Corollary 2. *Let $\overline{p}^{PR}$, $\overline{p}^{PS}$, and $\overline{p}^{AS}$ denote the optimal maximum verification intensities for pro-receiver, pro-sender, and anti-sender fact-checkers, respectively.*

- *In a sender-unfavorable environment, the anti-sender fact-checker never checks $\overline{p}^{AS} = 0$, while the pro-sender fact-checker checks more often than the pro-receiver one, $\overline{p}^{PS} \geq \overline{p}^{PR}$, with strict inequality whenever at least one of them uses imperfect verification.*
- *In a sender-favorable environment, the pro-sender fact-checker never checks $\overline{p}^{PS} = 0$, while the anti-sender fact-checker checks more often than the pro-receiver one, $\overline{p}^{AS} \geq \overline{p}^{PR}$, with strict inequality whenever at least one of them uses imperfect verification.*

An important implication of Corollary 2 is that the receiver may strictly prefer a partisan fact-checker over a neutral, pro-receiver one. What drives this result is a simple cardinal comparison: whenever the sender's payoff gain in the pivotal state exceeds the receiver's gain from learning the truth, the fact-checker who cares exclusively about the sender's preferences chooses a more aggressive policy. Specifically, the receiver prefers the pro-sender fact-checker in SUE and the anti-sender fact-checker in SFE precisely when the sender's incentive to manipulate the receiver's action is stronger than the receiver's own incentive to learn the truth.

5 Many fact-checkers

This section studies an extension of our baseline model in which the receiver has access to two fact-checkers, each choosing its own fact-checking policy.[18] Our objective is to

[18]When there are more than two fact-checkers, the equilibrium structure remains qualitatively the same.

characterize equilibrium verification intensities and to assess how multiple fact-checkers affect the overall provision of fact-checking.

Up until now, we assumed a single fact-checker. In reality, the receiver may have access to several fact-checking institutions, each with its own agenda. To study how this affects the provision of verification and the players' payoffs, suppose there are two fact-checkers with payoffs $u_{F,1}$ and $u_{F,2}$ and symmetric fact-checking technology, $c_1 = c_2 = c$. At the beginning of the game, each fact-checker decides on the fact-checking policies p_1, p_2. A non-silent message m is then checked with probability $p(m) := 1 - (1 - p_1(m))(1 - p_2(m))$. Then the game unfolds as in our baseline model. The sender observes the state θ and sends message m. The receiver sees sender's message m and realized fact-check outputs $(\mathcal{O}_1, \mathcal{O}_2)$ and then chooses a.[19]

The fact-checking policies chosen by the two fact-checkers jointly determine the overall verification probability for each non-silent message. The game then proceeds exactly as in the baseline model with fact-checking policy p with one of the corresponding p-equilibria characterized in Section 3 by the maximum verification intensity $\overline{p}$. For each fact-checker i, let $\overline{p}_i := \max\{p_i(0), p_i(1)\}$. To make predictions about the fact-checkers' choices of p_1 and p_2, we impose two assumptions on the selection of p-equilibria. First, among p-equilibria generating the same distribution of actions and states, only those that are not Pareto-dominated for the fact-checkers in terms of verification cost can be played.[20] Second, in SFE, we assume that the most informative equilibrium is selected, meaning that the 1-sender sends the most-checked message.[21] These assumptions guarantee that, after choosing their fact-checking policies, each fact-checker can anticipate the induced equilibrium outcome. A pair of policies (p_1, p_2) constitutes an equilibrium fact-checking profile if policies are mutual best responses given the continuation p-equilibrium.

[19]The informational content of two nonempty fact-check outputs is the same. Therefore, the receiver makes the same decision irrespective of whether she observed one or two nonempty fact-check outputs.

[20]We view this as a logical extension of the best-equilibrium selection in the case of one fact-checker.

[21]If we allow for a small fine for the sender that is caught in a lie, this extension would select the most informative equilibrium pattern. Interestingly, Nyhan and Reifler (2015) show that the speaker is less likely to receive negative fact-checking rating when fact-checking poses a salient threat in a form of reputational risks.

Before characterizing equilibrium fact-checking with two fact-checkers, note that if fact-checker i would choose no fact-checking when acting alone then the presence of another fact-checker cannot make checking more attractive since verification is a public good. In this case, its best response to any policy of the other fact-checker is simply $\overline{p}_i = 0$. Thus, strategic interaction is meaningful only in the case where both fact-checkers would choose a positive checking intensity if each were the unique fact-checker. We therefore focus on this case and characterize the equilibrium fact-checking policies.

Proposition 4. *Fix an environment. Suppose both fact-checkers have strictly positive gains in the relevant state, $\Delta_{+,i} = u_{F,i}(A,1) - u_{F,i}(R,1) > 0$ in SUE or $\Delta_{-,i} = u_{F,i}(R,0) - u_{F,i}(A,0) > 0$ in SFE, and face the same cost parameter c. There exist cost thresholds $\overline{c}$ and $\check{c}$, with $\overline{c} < \check{c}$, such that:*

- *For low fact-checking costs, $c \leq \overline{c}$, the only equilibria are asymmetric: one fact-checker never checks, while the other behaves as in the single fact-checker benchmark.*
- *For $c > \overline{c}$, there exists a unique interior equilibrium in which both fact-checkers check with strictly positive but imperfect intensities. In this equilibrium, each fact-checker's intensity $\overline{p}_i$ is increasing in its own gain and decreasing in the other fact-checker's gain.*
- *Let $\overline{p}^m$ denote the optimal verification intensity of the fact-checker with higher gains when operating alone. Then the overall equilibrium verification intensity $\overline{p}$ satisfies: (i) $\overline{p} = \overline{p}^m$ when $c \leq \overline{c}$; (ii) $\overline{p} < \overline{p}^m$ when $c \in (\overline{c}, \check{c})$; (iii) $\overline{p} > \overline{p}^m$ when $c > \check{c}$.*

When verification is cheap, each fact-checker wants the other to bear the cost of generating informative checks, so all equilibria are asymmetric: one fact-checker performs verification while the other free-rides and stays inactive. In this region, overall informativeness is the same as with a single fact-checker. When verification costs are moderate, neither fact-checker is willing to check as aggressively as it would when acting alone. Instead, each chooses a strictly positive but imperfect checking intensity, and their policies are strategic substitutes: a fact-checker checks less, the more aggressively the other checks. This pattern generates a unique interior equilibrium in which both check with positive but imperfect intensities, but in aggregate they verify less often than a single dedicated fact-checker would. Thus, a coordination problem leads to an underprovision of fact-checking. When verification becomes very costly, however, the joint verification effort of two strategic fact-checkers eventually exceeds what either would do alone. This is because each fact-checker now puts substantial weight on the possibility that the other may fail to verify, weakening free-riding incentives. Thus, the presence of multiple fact-checkers can either undermine or enhance informativeness depending purely on verification costs: it reduces verification when checks are valuable but moderately costly and strengthens verification only when claims are so difficult to verify that even a strongly motivated single fact-checker would check rarely.

Finally, we point out that the existence of the equilibrium with the underprovision of fact-checking relies on our assumption of the simultaneous fact-checkers' moves.[22] Moreover, our setting does not allow for repeated checks in case of the technology failure. We view both restrictions as reflecting the time-pressure conditions of real-world competition between fact-checking organizations. As pointed out by Graves (2016), "editors at FactCheck.org have remarked several times on the sharper deadline pressure the group faced once its national rivals appeared". FactCheck.org responded to new market conditions by introducing the "FactCheck Wire" in 2009 with the purpose of providing shorter fact-checks in a timely manner.

6 Discussion

This section considers two variations of our baseline model that allow us to discuss facts that can be checked by an extensive-margin fact-checking technology and are not binary in nature. We also discuss equilibrium selection.

Extensive-margin fact-checking. Our analysis assumes that the fact-checker chooses a verification intensity $p(m)$ for each message, which determines the probability that a check produces a conclusive output. An alternative modeling choice is to assume that the fact-checker instead chooses whether to check a message at a cost, and that a checked

[22] If the fact-checkers moved sequentially, then the first fact-checker would have a first-mover advantage adopting a no fact-checking policy, passing the need to check to the second-mover.

message is verified perfectly but with an exogenous probability of failure. We can show that this extensive-margin formulation leads to fact-checking policies that are "bang–bang" in nature: the optimal policy is either full checking or no checking, and the receiver's payoff depends entirely on whether the cost crosses a cost threshold. By contrast, in the intensity-based framework developed in this paper, the fact-checker optimally selects interior verification intensities whenever verification is sufficiently costly, leading to richer comparative statics and welfare implications. Interior verification levels also allow for smooth substitution across fact-checkers when multiple fact-checkers are present. Thus, while both frameworks capture technological frictions in fact-checking, the intensity-based technology generates more nuanced predictions about optimal fact-checking, comparative statics, and strategic interaction.

Larger state space. Our model considers only claims about the binary issues. In practice, fact-checkers check variety of statements, some of them quantitative in nature.[23] One way to allow for such statements is to enlarge the state space, so that $\theta \in [0,1]$. For simplicity, suppose that the prior is uniform on $[0,1]$ and the message space may contain only the closed intervals that are subsets of the unit interval. For such state space, Titova and Zhang (2025) show that the sender can achieve the commitment outcome in SUE with verifiable information only.[24] The solution involves a winning message $m_w = [\theta^*, 1]$ and a losing message $m_l = [0, \theta^*]$, where the cutoff value θ^* is chosen to make the receiver exactly indifferent between taking actions $a = A$ and $a = R$ upon observing m_w. The tie is broken in the sender's favor. In our setting, messages are cheap but can be verified at a cost determined by the chosen fact-checking intensity. Thus, the pro-sender fact-checker is able to deliver the sender's commitment payoff in SUE, if the fact-checking cost is low enough. In particular, the fact-checker only checks m_w with probability one. The outcome does not rely on the selection and does not involve randomization on the sender's side. In our binary setting, the commitment payoff is not achievable, since it requires undetectable

[23] For example, Donald Trump famously spread information about US unemployment rates that received negative fact-checking ratings (National Public Radio, 2017).

[24] The definitions of SUE and SFE remain the same. In SUE (SFE), the receiver rejects (accepts) the sender's proposal under the prior.

randomization by the 0-sender which the fact-checker cannot sustain without revealing him. Note that the anti-sender fact-checker uses a similar structure to implement the sender-worst outcome in SFE, with a difference that the cutoff value θ^* for messages m_w and m_l is chosen to make the receiver exactly indifferent between taking actions $a = A$ and $a = R$ upon observing m_l and the tie is broken against the sender.

Equilibrium selection. We assume that the fact-checker can steer the sender and the receiver toward its favorite p-equilibrium. We now briefly discuss how our results change under the opposite assumption of the worst p-equilibrium selection. In SUE, equilibrium outcomes are uniquely pinned down by the maximum verification intensity. The fact-checker's benefit from checking is hence unaffected by equilibrium selection but the effective cost of implementing $\overline{p}$ rises, since both sender types pool on the most-checked message. This decreases a cost threshold for optimal full checking but leaves the qualitative structure of the optimal policy intact. In SFE, a pooling-on-silence equilibrium always exists, generates no information, and defeats any attempt by the fact-checker to induce informative communication. Thus, in SFE under the worst-equilibrium selection, the fact-checker never checks.

7 Concluding remarks

This paper highlights how the informational role of fact-checkers is shaped by their incentives and by the strategic environment in which they operate. We show that strategic fact-checkers intervene only when verification can favorably correct the decision-making, and that the receiver may benefit more from partisan fact-checkers than from neutral ones. When multiple fact-checkers interact, free-riding generates a coordination problem that depresses verification relative to a single dedicated fact-checker, unless fact-checking is sufficiently costly. More broadly, our analysis illustrates how the value of fact-checking arises not merely from detecting falsehoods, but from how verification reshapes strategic communication. These insights underscore the importance of understanding fact-checkers' objectives as automated and institutional fact-checking continue to expand.

References

ALLCOTT, HUNT and MATTHEW GENTZKOW (2017), “Social Media and Fake News in the 2016 Election”, *Journal of Economic Perspectives*, 31, 2, pp. 211-36.

AMAZEEN, MICHELLE A. (2015), “Revisiting the Epistemology of Fact-Checking”, *Critical Review*, 27, 1, pp. 1-22.

— (2016), “Checking the Fact-Checkers in 2008: Predicting Political Ad Scrutiny and Assessing Consistency”, *Journal of Political Marketing*, 15, 4, pp. 433-464.

ANDREOTTOLA, GIOVANNI and CHARLES LOUIS-SIDOIS (2025), “Why Us and not Them? A Theory of Political Fact-Checking”, *Working Paper*.

BALBUZANOV, IVAN (2019), “Lies and Consequences”, *International Journal of Game Theory*, 48, 4, pp. 1203-1240.

BARON, DAVID P. and DAVID BESANKO (1984), “Regulation, Asymmetric Information, and Auditing”, *The RAND Journal of Economics*, pp. 447-470.

BARRERA, OSCAR, SERGEI GURIEV, EMERIC HENRY, and EKATERINA ZHURAVSKAYA (2020), “Facts, Alternative Facts, and Fact Checking in Times of Post-Truth Politics”, *Journal of Public Economics*, 182, p. 104123.

BORDER, KIM C. and JOEL SOBEL (1987), “Samurai Accountant: A Theory of Auditing and Plunder”, *The Review of Economic Studies*, 54, 4, pp. 525-540.

DZIUDA, WIOLETTA and CHRISTIAN SALAS (2018), “Communication with Detectable Deceit”, *Working Paper*.

EDERER, FLORIAN and WEICHENG MIN (2024), “Bayesian Persuasion with Lie Detection”, *Working Paper*.

GRAVES, LUCAS (2016), *Deciding What’s True: The Rise of Political Fact-Checking in American Journalism*, Columbia University Press, p. 336.

— (2017), “Anatomy of a Fact Check: Objective Practice and the Contested Epistemology of Fact Checking”, *Communication, Culture & Critique*, 10, 3, pp. 518-537.

— (2018), “Understanding the Promise and Limits of Automated Fact-Checking”, *Reuters Institute for the Study of Journalism*.

GRAVES, LUCAS and FEDERICA CHERUBINI (2016), “The Rise of Fact-Checking Sites in Europe”, *Reuters Institute for the Study of Journalism*.

Guo, Yingni and Eran Shmaya (2021), “Costly Miscalibration”, *Theoretical Economics*, 16, 2, pp. 477-506.

Hassan, Naeemul, Fatma Arslan, Chengkai Li, and Mark Tremayne (2017), “Toward Automated Fact-Checking: Detecting Check-Worthy Factual Claims by Claimbuster”, *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 1803-1812.

Holm, Håkan J. (2010), “Truth and Lie Detection in Bluffing”, *Journal of Economic Behavior & Organization*, 76, 2, pp. 318-324.

Ivanov, Maxim (2010), “Communication via a Strategic Mediator”, *Journal of Economic Theory*, 145, 2, pp. 869-884.

Kamenica, Emir and Matthew Gentzkow (2011), “Bayesian Persuasion”, *American Economic Review*, 101, 6 (Oct. 2011), pp. 2590-2615.

Kolotilin, Anton, Tymofiy Mylovanov, Andriy Zapechelnyuk, and Ming Li (2017), “Persuasion of a Privately Informed Receiver”, *Econometrica*, 85, 6, pp. 1949-1964.

Laffont, Jean-Jacques and Jean Tirole (1986), “Using Cost Observation to Regulate Firms”, *Journal of political Economy*, 94, 3, Part 1, pp. 614-641.

Lim, Chloe (2018), “Checking how Fact-Checkers Check”, *Research & Politics*, 5, 3.

Louis-Sidois, Charles (2025), “Both Judge and Party? Investigating the Political Unbiasedness of Fact-checkers”, *Journal of the European Economic Association*, jvaf011.

Marietta, Morgan, David C. Barker, and Todd Bowser (2015), “Fact-checking Polarized Politics: Does the Fact-check Industry Provide Consistent Guidance on Disputed Realities?”, *The Forum*, 4, De Gruyter, vol. 13, pp. 577-596.

Mathevet, Laurent, Jacopo Perego, and Ina Taneva (2020), “On Information Design in Games”, *Journal of Political Economy*, 128, 4, pp. 1370-1404.

Mookherjee, Dilip and Ivan Png (1989), “Optimal Auditing, Insurance, and Redistribution”, *The Quarterly Journal of Economics*, 104, 2, pp. 399-415.

National Public Radio (2017), *Ahead Of Trump’s First Jobs Report, A Look At His Remarks On The Numbers*, `https://www.npr.org/2017/01/29/511493685/ahead-of-trumps-first-jobs-report-a-look-at-his-remarks-on-the-numbers`, Last accessed on 2021-12-30.

Nieminen, Sakari and Lauri Rapeli (2019), “Fighting Misperceptions and Doubting Journalists’ Objectivity: A Review of Fact-Checking Literature”, *Political Studies Review*, 17, 3, pp. 296-309.

NYHAN, BRENDAN, ETHAN PORTER, JASON REIFLER, and THOMAS WOOD (2020), “Taking Fact-checks Literally but not Seriously? The Effects of Journalistic Fact-Checking on Factual Beliefs and Candidate Favorability”, *Political Behavior*, 42, 3, pp. 939-960.

NYHAN, BRENDAN and JASON REIFLER (2010), “When Corrections Fail: The Persistence of Political Misperceptions”, *Political Behavior*, 32, 2, pp. 303-330.

— (2015), “The Effect of Fact-checking on Elites: A Field Experiment on US State Legislators”, *American Journal of Political Science*, 59, 3, pp. 628-640.

OSTERMEIER, ERIC (2011), “Selection Bias? PolitiFact Rates Republican Statements as False at Three Times the Rate of Democrats”, *Smart Politics*, 10.

SADAKANE, HITOSHI and YIN CHI TAM (2022), “Cheap talk and Lie detection”, *Working Paper*.

SALAMANCA, ANDRÉS (2021), “The Value of Mediated Communication”, *Journal of Economic Theory*, 192, p. 105191.

SHISHKIN, DENIS (2025), “Evidence Design and Voluntary Disclosure”, *Working Paper*.

TITOVA, MARIA and KUN ZHANG (2025), “Persuasion with Verifiable Information”, *Working Paper*.

TOWNSEND, ROBERT M. (1979), “Optimal Contracts and Competitive Markets with Costly State Verification”, *Journal of Economic theory*, 21, 2, pp. 265-293.

USCINSKI, JOSEPH (2015), “The Epistemology of Fact Checking (is Still Naive): Rejoinder to Amazeen”, *Critical Review*, 27, 2, pp. 243-252.

USCINSKI, JOSEPH and RYDEN BUTLER (2013), “The Epistemology of Fact Checking”, *Critical Review*, 25, 2, pp. 162-180.

WALTER, NATHAN, JONATHAN COHEN, R. LANCE HOLBERT, and YASMIN MORAG (2020), “Fact-Checking: A Meta-Analysis of What Works and for Whom”, *Political Communication*, 37, 3, pp. 350-375.

WEEKS, BRIAN E. (2015), “Emotions, Partisanship, and Misperceptions: How Anger and Anxiety Moderate the Effect of Partisan Bias on Susceptibility to Political Misinformation”, *Journal of Communication*, 65, 4, pp. 699-719.

WEEKS, BRIAN E. and R. KELLY GARRETT (2014), “Electoral Consequences of Political Rumors: Motivated Reasoning, Candidate Rumors, and Vote Choice during the 2008 US Presidential Election”, *International Journal of Public Opinion Research*, 26, 4, pp. 401-422.

WINTERSIECK, AMANDA, KIM FRIDKIN, and PATRICK KENNEY (2021), “The Message Matters: The Influence of Fact-Checking on Evaluations of Political Messages”, *Journal of Political Marketing*, 20, 2, pp. 93-120.

Appendix

Proof of Proposition 1

Fix the maximum verification intensity $\overline{p} = \max\{p(0), p(1)\}$ of some fact-checking policy p. Let (σ, α, π) be a p-equilibrium. For any non-silent $m \in \{0,1\}$, consistency with the fact-checking technology implies $\pi(m, \mathcal{T}) = \mathbf{1}\{m = 1\}$ and $\pi(m, \mathcal{F}) = \mathbf{1}\{m = 0\}$. When $\theta = 1$, any conclusive outcome reveals $\theta = 1$, and when $\theta = 0$, any conclusive outcome reveals $\theta = 0$.

We start from SUE: $\mu < \omega$. First, we show that in any p-equilibrium we have $U_S(0) = 0$. Suppose towards the contrary that $U_S(0) > 0$. Then there exists an on-path message $m \in \mathcal{M}$ such that $\sigma(m|0) > 0$ and the 0-sender is sometimes accepted after sending m. Since any conclusive fact-check output reveals $\theta = 0$, the receiver strictly prefers to reject after such an output. Therefore, acceptance for the 0-sender can only occur when the outcome is inconclusive, $\mathcal{O} = \varnothing$. This implies that for some on-path message m, $\sigma(m|0) > 0$, $p(m) < 1$, and $\alpha(m, \varnothing) > 0$.

The receiver's posterior belief after observing $(m, \varnothing)$ is

$$\pi(m, \varnothing) = \frac{\mu\sigma(m|1)}{\mu\sigma(m|1) + (1-\mu)\sigma(m|0)} \, .$$

Since $\alpha(m, \varnothing) > 0$ and the receiver's strategy must be optimal, we must have $\pi(m, \varnothing) \geq \omega$. This inequality is equivalent to

$$\sigma(m|1) \geq \frac{1-\mu}{\mu} \cdot \frac{\omega}{1-\omega} \cdot \sigma(m|0) \ > \sigma(m|0) \, ,$$

where the second inequality follows from the SUE condition.

Let $\mathcal{M}^+$ be the set of on-path messages m such that $\sigma(m|0) > 0$, $p(m) < 1$, and $\alpha(m, \varnothing) > 0$. Thus, we have $\sigma(m|1) > \sigma(m|0)$ for every $m \in \mathcal{M}^+$, and $\sum_{m \in \mathcal{M}^+} \sigma(m|1) > \sum_{m \in \mathcal{M}^+} \sigma(m|0)$. Then there exists $m' \notin \mathcal{M}^+$ with $\sigma(m'|0) > 0$, such that $p(m') = 1$ or $\alpha(m', \varnothing) = 0$. In both cases, 0-sender is not accepted after $(m', \varnothing)$. Then 0-sender has a profitable deviation from ever sending m' and we arrive at a contradiction. Thus, $U_S(0) = 0$ in any p-equilibrium.

Now we show that $U_S(1) = \overline{p}$. Note that 1-sender can only be rejected when the fact-check output is inconclusive, $\mathcal{O} = \varnothing$. From the previous argument for 0-sender, the receiver must reject after any on-path inconclusive output in p-equilibrium, $\alpha(m,\varnothing)=0$. As a result, the acceptance probability of 1-sender is given by the probability that a conclusive output occurs: $U_S(1)=\sum_{m\in\mathcal{M}}\sigma(m|1)p(m)$. Furthermore, by the optimality for 1-sender, it has to be the case that $p(m)>p(m')$ implies $\sigma(m'|1)=0$. As a result, $U_S(1)=\overline{p}$.

An example of p-equilibrium with these payoffs is as follows. The sender's strategy is $\sigma(\overline{m}|1)=\sigma(\overline{m}|0)=1$. The receiver's acceptance strategy is $\alpha(m,\mathcal{T})=1$, $\alpha(m,\mathcal{F})=\alpha(m,\varnothing)=0$ for any m. Finally, $\pi(m,\varnothing)<\omega$ for any m. This is clearly p-equilibrium as no sender's type has a profitable deviation.

We now turn to the sender-favorable environment, $\mu>\omega$. We first show that in any p-equilibrium we have $U_S(1)=1$. Suppose towards the contrary that $U_S(1)<1$. Then 1-sender is sometimes rejected in p-equilibrium, which can only occur after an inconclusive outcome, $\mathcal{O}=\varnothing$. Thus, there exists an on-path message $m\in\mathcal{M}$ such that $\sigma(m|1)>0$, $p(m)<1$, and $\alpha(m,\varnothing)<1$.

The receiver's posterior belief after observing $(m,\varnothing)$ is

$$\pi(m,\varnothing)=\frac{\mu\sigma(m|1)}{\mu\sigma(m|1)+(1-\mu)\sigma(m|0)}\,.$$

Since $\alpha(m,\varnothing)<1$ and the receiver's strategy must be optimal, we must have $\pi(m,\varnothing)\leq\omega$. This inequality is equivalent to

$$\sigma(m|0)\geq\frac{\mu}{1-\mu}\cdot\frac{1-\omega}{\omega}\cdot\sigma(m|1)>\sigma(m|1),$$

where the second inequality follows from SFE condition.

Let $\mathcal{M}^-$ be the set of on-path messages m such that $\sigma(m|1)>0$, $p(m)<1$, and $\alpha(m,\varnothing)<1$. Thus, we have $\sigma(m|1)<\sigma(m|0)$ for every $m\in\mathcal{M}^-$, and $\sum_{m\in\mathcal{M}^-}\sigma(m|1)<\sum_{m\in\mathcal{M}^-}\sigma(m|0)$. Then there exists $m'\notin\mathcal{M}^-$ with $\sigma(m'|1)>0$, such that $p(m')=1$ or $\alpha(m',\varnothing)=1$. In both cases, 1-sender gets accepted, and this remains true for any $m'\notin\mathcal{M}^-$. As a result, 1-sender always gets accepted, and we arrive at a contradiction. Hence, $U_S(1)=1$ in any p-equilibrium.

We now bound the 0-sender's payoff: $U_S(0)\geq 1-\overline{p}$ in any p-equilibrium. Since $U_S(1)=1$, type 1 must be accepted with probability one. This implies that for any on-path message m with $\sigma(m|1)>0$ and $p(m)<1$, the receiver must also accept after the inconclusive output, $\alpha(m,\varnothing) = 1$. Pick any such on-path message m with $\sigma(m|1) > 0$. If $p(m) = 1$, then there is nothing to prove. If $p(m) < 1$, the 0-sender can deviate to always sending m and get accepted whenever the check fails, with probability $1 - p(m)$. Thus, 0-sender can secure a payoff of at least $1 - p(m)$, and thus in any p-equilibrium we have $U_S(0) \geq 1 - \max_m p(m) = 1 - \overline{p}$.

Finally, we construct p-equilibrium that delivers any $U_S(0) \in [1 - \overline{p}, 1]$ in SFE. The sender's strategy is $\sigma(\overline{m}|1) = q$, $\sigma(\underline{m}|1) = 1 - q$ and $\sigma(\underline{m}|0) = 1$, with $q < \frac{\mu - \omega}{\mu(1-\omega)}$. The receiver's acceptance strategy is $\alpha(m, \mathcal{T}) = 1$, $\alpha(m, \mathcal{F}) = 0$ for any m, and $\alpha(\overline{m}, \varnothing) = \alpha(\underline{m}, \varnothing) = 1$, $\alpha(m_s, \varnothing) = 0$. Finally, $\pi(\overline{m}, \varnothing) = 1$, $\pi(\underline{m}, \varnothing) = \frac{\mu(1-q)}{\mu(1-q)+(1-\mu)}$, and $\pi(m_s, \varnothing) < \omega$. This is p-equilibrium, where 0-sender sends the least-checked on-path message, and 1-sender always gets the proposal accepted. Note that $U_S(0) = 1 - \underline{p}$. Thus, any $U_S(0) \in [1 - \overline{p}, 1]$ can be attained by moving $\underline{p} \in [0, \overline{p}]$.

Proof of Proposition 2

Fix the maximum verification intensity $\overline{p} = \max\{p(0), p(1)\}$ for fact-checking policies p and an environment (μ, ω).

From Proposition 1, we have $U_S(0) = 0$ and $U_S(1) = \overline{p}$ in SUE. Acceptance can occur only after a conclusive fact-check revealing $\theta = 1$. Otherwise, the receiver rejects. Thus, the receiver accepts if and only if $\theta = 1$ and the fact-check is conclusive. From the proof of Proposition 1, this event occurs with probability $\mu\overline{p}$. Conditional on acceptance in state $\theta = 1$, the receiver's payoff is $1 - \omega$, and it is zero otherwise. Hence, $U_R = \mu(1 - \omega)\overline{p}$. This payoff is unique for a given $\overline{p}$. Together with $U_S = \mu\overline{p}$, this establishes the SUE part of the proposition.

From Proposition 1, we have $U_S(0) \in [1 - \overline{p}, 1]$ and $U_S(1) = 1$ in SFE. If the receiver accepts under the prior, her expected payoff is $\mu - \omega$, which is the *no-communication payoff* of the receiver. Fact-checking can only benefit the receiver by preventing regrettable acceptance when $\theta = 0$. Each time $\theta = 0$ and a conclusive fact-check occurs, the receiver avoids an error of size ω. Thus, relative to the no-communication payoff, the benefit is $(1 - \mu)p_v\omega$, where p_v denotes the verification probability of the messages the 0-sender uses on-path. In p-equilibrium, the 0-sender's payoff is $U_S(0) = 1 - p_v$. From Proposition 1, $p_v \in [0, \overline{p}]$, and every value in this interval is attainable in some p-equilibrium of some fact-checking policy p with the maximum verification intensity $\overline{p}$. Therefore, the receiver's payoff is $U_R = \mu - \omega + (1-\mu)\omega p_v$. As $p_v \in [0, \overline{p}]$, $U_R \in [\mu - \omega, \mu - \omega + (1-\mu)\omega\overline{p}]$. These bounds are attainable by the equilibria constructed in the proof of Proposition 1. The lower bound arises when the 0-sender uses only unchecked messages ($p_v = 0$), and the upper bound arises when the 0-sender uses only maximally checked messages ($p_v = \overline{p}$). Interior receiver's payoff arises when the 0-sender uses only $\underline{m}$ ($p_v = \underline{p}$). Together with $U_S \in [1 - (1-\mu)\overline{p}, 1]$, this establishes the SFE part of the proposition.

Proof of Lemma 1

Fix an environment (μ, ω). There is nothing to prove for $\overline{p} = 0$. First, fix maximum verification intensity $\overline{p} \in (0, 1)$.

From Proposition 1, in SUE any p-equilibrium with maximum verification intensity $\overline{p}$ has $U_S(0) = 0$ and $U_S(1) = \overline{p}$. To attain $U_S(1) = \overline{p}$, it has to be the case that $\sigma(\overline{m}|1) = 1$. For the 0-sender to get payoff zero, he must be always rejected. In particular, after an inconclusive outcome following $\overline{m}$, the receiver must reject. Let $\sigma(\overline{m}|0) = q > 0$. The receiver's posterior belief after observing $\overline{m}$ and the inconclusive fact-check output $\varnothing$ is

$$\pi(m, \varnothing) = \frac{\mu}{\mu + (1-\mu)q} \, .$$

Rejection after an inconclusive outcome requires $\pi(\overline{m}, \varnothing) \leq \omega$, which is equivalent to

$$q \geq \frac{\mu(1-\omega)}{\omega(1-\mu)} \, .$$

Thus, in any implementation of $\overline{p} < 1$, the 0-sender must put at least $\frac{\mu(1-\omega)}{\omega(1-\mu)}$ probability on $\overline{m}$. Thus, the cheapest p-equilibrium has $\sigma(\overline{m}|0) = \frac{\mu(1-\omega)}{\omega(1-\mu)}$ and $\sigma(m_s|0) = 1 - \sigma(\overline{m}|0)$. As a result,

$$K(\overline{p}) = \left[\mu + (1-\mu)\frac{\mu(1-\omega)}{\omega(1-\mu)}\right] C(\overline{p}) = \frac{\mu}{\omega} C(\overline{p}) \, .$$

From Proposition 1, in SFE any p-equilibrium with maximum verification intensity $\overline{p}$ has $U_S(1) = 1$ and $U_S(0) \in [1 - \overline{p}, 1]$. To implement $\overline{p} < 1$, the fact-checker wants the 0-sender to face $\overline{p}$ whenever he pools with the 1-sender. The cheapest and the only way to do this is to have both types always pool on $\overline{m}$. Because $\overline{m}$ is sent with probability one in this equilibrium, the expected cost is $K(\overline{p}) = C(\overline{p})$.

Now fix $\overline{p} = 1$. In both environments, there is a low-cost separating equilibrium: the 1-sender always sends $\overline{m}$ and is verified truthful and accepted, while the 0-sender remains silent and is rejected. Then $\overline{m}$ is sent with probability μ, and the expected cost is $K(1) = \mu C(1)$. Any implementation in which the 0-sender also uses a checked non-silent message raises the frequency of costly checks above μ and thus costs more than $\mu C(1)$.

Proof of Proposition 3

Consider SUE, $\mu < \omega$. From Proposition 1, the induced distribution of actions and states $\lambda(a,\theta|\overline{p})$ satisfies $\lambda(A,1|\overline{p}) = \mu\overline{p}$, $\lambda(R,1|\overline{p}) = \mu(1-\overline{p})$, and $\lambda(R,0|\overline{p}) = 1-\mu$. By Lemma 1, we then can write the fact-checker's problem as

$$\mu\overline{p}u_F(A,1) + \mu(1-\overline{p})u_F(R,1) + (1-\mu)u_F(R,0) - K(\overline{p}) \to \max_{\overline{p}\in[0,1]} .$$

Denote the portion of the fact-checker's objective function that depends on $\overline{p}$ as $W(\overline{p}) = \mu\Delta_+\overline{p} - K(\overline{p})$. Clearly, when $\Delta_+ \leq 0$, $\overline{p}^* = 0$ is optimal. In what follows, we consider $\Delta_+ > 0$. For $\overline{p} \in [0,1)$, we have $W(\overline{p}) = \mu\Delta_+\overline{p} - \frac{\mu}{\omega}\frac{c\overline{p}^2}{2}$. If this function has an interior maximizer, then it is given by $\overline{p}^* = \frac{\omega\Delta_+}{c}$, achieving the value $W(\overline{p}^*) = \mu\omega\frac{\Delta_+^2}{2c}$. This requires $c > \omega\Delta_+$. For $\overline{p} = 1$, we have $W(1) = \mu\Delta_+ - \mu\frac{c}{2}$. We have $W(\overline{p}^*) > W(1)$ if and only if $c < (1-\sqrt{1-\omega})\Delta_+$ or $c > (1+\sqrt{1-\omega})\Delta_+$. Combining these inequalities with $c > \omega\Delta_+$, we get the interior optimum $\overline{p}^* = \frac{\omega\Delta_+}{c}$ if and only if $c > (1+\sqrt{1-\omega})\Delta_+$. Otherwise, $\overline{p}^* = 1$.

Consider SFE, $\mu > \omega$. From Proposition 1, the induced distribution of actions and states $\lambda(a,\theta|\overline{p})$ satisfies $\lambda(A,1|\overline{p}) = \mu$, $\lambda(A,0|\overline{p}) \in [(1-\mu)(1-\overline{p}),(1-\mu)]$, and $\lambda(R,0|\overline{p}) \in [0,(1-\mu)\overline{p}]$. Due to the linearity of the expected fact-checker's benefit, if the fact-checker wants to check sometimes, then it will find it optimal to implement $\overline{p}$. Thus, $\lambda(A,1|\overline{p}) = \mu$, $\lambda(A,0|\overline{p}) = (1-\mu)(1-\overline{p})$, and $\lambda(R,0|\overline{p}) = (1-\mu)\overline{p}$. By Lemma 1, we then can write the fact-checker's problem as

$$\mu u_F(A,1) + (1-\mu)(1-\overline{p})u_F(A,0) + (1-\mu)\overline{p}u_F(R,0) - K(\overline{p}) \to \max_{\overline{p}\in[0,1]} .$$

Denote the portion of the fact-checker's objective function that depends on $\overline{p}$ as $V(\overline{p}) = (1-\mu)\Delta_-\overline{p} - K(\overline{p})$. Clearly, when $\Delta_- \leq 0$, $\overline{p}^* = 0$ is optimal. In what follows, we consider $\Delta_- > 0$. For $\overline{p} \in [0,1)$, we have $V(\overline{p}) = (1-\mu)\Delta_-\overline{p} - \frac{c\overline{p}^2}{2}$. If this function has an interior maximizer, then it is given by $\overline{p}^* = \frac{(1-\mu)\Delta_-}{c}$, achieving the value $V(\overline{p}^*) = (1-\mu)^2\frac{\Delta_-^2}{2c}$. This requires $c > (1-\mu)\Delta_-$. For $\overline{p} = 1$, we have $V(1) = (1-\mu)\Delta_- - \mu\frac{c}{2}$. We have $V(\overline{p}^*) > V(1)$ if and only if $c < \frac{(1-\mu)\Delta_-}{1+\sqrt{1-\mu}}$ or $c > \frac{(1-\mu)\Delta_-}{1-\sqrt{1-\mu}}$. Combining these inequalities with $c > (1-\mu)\Delta_-$, we get the interior optimum $\overline{p}^* = \frac{(1-\mu)\Delta_-}{c}$ if and only if $c > \frac{(1-\mu)\Delta_-}{1-\sqrt{1-\mu}} = \frac{(1-\mu)\Delta_-}{\mu}(1+\sqrt{1-\mu})$. Otherwise, $\overline{p}^* = 1$.

Proof of Corollary 1

Since $u_R(A,1) = 1-\omega$, $u_R(R,1) = 0$, $u_R(A,0) = -\omega$, and $u_R(R,0) = 0$ for the receiver and $u_S(A) = 1$ and $u_S(R) = 0$ for the sender, we have $\Delta_+ = u_F(A,1) - u_F(R,1) = \beta_S + \beta_R(1-\omega)$ and $\Delta_- = u_F(R,0) - u_F(A,0) = -\beta_S + \beta_R\omega$. By Proposition 2, the receiver's ex ante payoff is increasing in the maximum verification intensity $\overline{p}$ in both environments, uniquely in SUE and in a strong-order set sense in SFE. Thus, the receiver weakly benefits under fixed p-equilibrium selection from greater $\overline{p}$. By Proposition 3, optimal $\overline{p}$ weakly increases in Δ_+ in SUE. As a result, increasing either β_S or β_R increases optimal $\overline{p}$. By Proposition 3, optimal $\overline{p}$ also increases in Δ_- in SFE. As a result, decreasing either β_S or increasing β_R increases optimal $\overline{p}$.

Proof of Corollary 2

For the pro-receiver fact-checker, we have $\Delta_+^{PR} = 1-\omega$ and $\Delta_-^{PR} = \omega$. For the pro-sender fact-checker, we have $\Delta_+^{PS} = 1$ and $\Delta_-^{PS} = -1$. For the anti-sender fact-checker, we have $\Delta_+^{AS} = -1$ and $\Delta_-^{AS} = 1$. The result then follows, as $\overline{p}$ weakly increases in Δ_+ in SUE and in Δ_- in SFE, with a strict increase in the interior region.

Proof of Proposition 4

First, the best response of the fact-checker i to $\overline{p}_j = 1$ is $\overline{p}_i = 0$. Now suppose the fact-checker i believes that $\overline{p}_j \in [0,1)$.

Consider SUE, $\mu < \omega$. To simplify notation, denote $\Delta_{+,i}$ as Δ_i. Without loss of generality, suppose $\Delta_1 \geq \Delta_2$. Further, denote the "monopoly" cost threshold of fact-checker i as $\overline{c}_i = c_{SUE}(u_{F,i}) = \Delta_i(1+\sqrt{1-\omega})$. Following the steps of derivation of the optimal fact-checking in Proposition 3 (changing from variable Δ_+ to $(1-\overline{p}_j)\Delta_i$), the best response of the fact-checker i is

$$\overline{p}_i = \begin{cases} 1\,, & \text{if } 0 < c \leq (1-\overline{p}_j)\overline{c}_i\,, \\ \dfrac{\omega(1-\overline{p}_j)\Delta_i}{c}\,, & \text{if } c > (1-\overline{p}_j)\overline{c}_i\,. \end{cases}$$

As a result, when c is sufficiently high, we have an equilibrium in which

$$\overline{p}_i = \frac{\omega\Delta_i(c-\omega\Delta_j)}{c^2-\omega^2\Delta_i\Delta_j}\,.$$

We first note that the interior equilibrium requires $c > \omega\Delta_1 \geq \omega\Delta_2$, as otherwise we cannot get the equilibrium in which both $\overline{p}_1$ and $\overline{p}_2$ are strictly positive. We also get $\frac{\partial \overline{p}_i}{\partial \Delta_i} > 0$ and $\frac{\partial \overline{p}_i}{\partial \Delta_j} = -\frac{\omega^2\Delta_i c(c-\omega\Delta_i)}{(c^2-\omega^2\Delta_i\Delta_j)^2} < 0$. The exact condition of sufficiently high cost arises from satisfying $c > (1-\overline{p}_j)\overline{c}_i$. Using explicit expressions for $\overline{p}_1$ and $\overline{p}_2$, we can show that these conditions are equivalent to requiring

$$c > \frac{\overline{c}_1 + \sqrt{\overline{c}_1^2 - \overline{c}_1\overline{c}_2\kappa(\omega)}}{2} := \overline{c} \geq \omega\Delta_1\,,$$

where $\kappa(\omega) = \frac{4\omega(1-\omega+\sqrt{1-\omega})}{(1+\sqrt{1-\omega})^2}$ is hump-shaped with $\kappa(0) = \kappa(1) = 0$ and achieves maximum at $\omega = \frac{3}{4}$, with $\kappa\left(\frac{3}{4}\right) = 1$. As a result, we get the interior equilibrium above if and only if $c > \overline{c}$. Otherwise, the only equilibria are in which one of $\overline{p}_1$, $\overline{p}_2$ is given by Proposition 3, while the other is zero.

Note that $\overline{c} < \overline{c}_1$. As a result, when $c \in (\overline{c}, \overline{c}_1]$, the joint maximum verification intensity of the interior equilibrium is strictly below the intensity of one achieved by the sole fact-checker 1. Now we consider $c > \overline{c}_1$. Using explicit expressions for $\overline{p}_1$ and $\overline{p}_2$, we get

$$\overline{p} = 1 - (1-\overline{p}_1)(1-\overline{p}_2) = 1 - \frac{c^2(c-\omega\Delta_1)(c-\omega\Delta_2)}{(c^2-\omega^2\Delta_1\Delta_2)^2}\,.$$

Thus, we obtain

$$\frac{\partial \overline{p}}{\partial \Delta_i} = \omega c^2 (c - \omega \Delta_j) \frac{c^2 - 2\omega \Delta_j c + \omega^2 \Delta_i \Delta_j}{(c^2 - \omega^2 \Delta_i \Delta_j)^3} \, .$$

Thus, $\frac{\partial \overline{p}}{\partial \Delta_1} > 0$ and $\overline{p}$ is either U-shaped or increasing in $\Delta_2 \in [0, \Delta_1]$, since $c^2 - 2\omega \Delta_1 c + \omega^2 \Delta_1 \Delta_2$ is increasing in Δ_2 and strictly positive when $\Delta_2 = \Delta_1$. Furthermore,

$$\overline{p}(\Delta_1, \Delta_2 = 0) = \frac{\omega \Delta_1}{c} \, ,$$

and

$$\overline{p}(\Delta_1, \Delta_2 = \Delta_1) = \frac{\omega \Delta_1 (2c + \omega \Delta_1)}{(c + \omega \Delta_1)^2} \, .$$

As a result, whenever the interior equilibrium exists and $c < \frac{1+\sqrt{5}}{2} \omega \Delta_1$, the joint maximum verification intensity in the two-checker interior equilibrium is strictly below the intensity chosen by fact-checker 1 when operating alone, $\overline{p}(\Delta_1, \Delta_2 = 0)$. Finally, we note that $\overline{p}(\Delta_1, \Delta_2) > \overline{p}(\Delta_1, \Delta_2 = 0)$ if and only if $c^4 - 3\omega \Delta_1 c^3 + 2\omega^2 \Delta_1^2 c^2 + \omega^3 \Delta_1^2 \Delta_2 c - \omega^4 \Delta_1^3 \Delta_2 > 0$, or $(c - \omega \Delta_1)(c^3 - 2\omega \Delta_1 c^2 + \Delta_1^2 \Delta_2 \omega^3) > 0$, or $c^3 - 2\omega \Delta_1 c^2 + \Delta_1^2 \Delta_2 \omega^3 > 0$. This cubic inequality is satisfied for sufficiently large cost $c > c^{\dagger}$. As a result, when $c > \max\{\overline{c}_1, c^{\dagger}\}$, we conclude that $\overline{p}$ strictly exceeds $\overline{p}(\Delta_1, \Delta_2 = 0)$. This completes the proof for SUE.

Consider SFE, $\mu > \omega$. With a slight abuse of notation, denote $\Delta_{-,i}$ as Δ_i, suppose $\Delta_1 \geq \Delta_2$, and denote the "monopoly" cost threshold of fact-checker i as $\overline{c}_i = c_{SFE}(u_{F,i}) = \frac{1-\mu}{\mu} \Delta_i (1 + \sqrt{1-\mu})$. Following the steps of derivation of the optimal fact-checking in Proposition 3 (changing from variable Δ_- to $(1 - \overline{p}_j)\Delta_i$), the best response of the fact-checker i is

$$\overline{p}_i = \begin{cases} 1 \, , & \text{if } 0 < c \leq (1 - \overline{p}_j)\overline{c}_i \, , \\ \dfrac{(1-\mu)(1-\overline{p}_j)\Delta_i}{c} \, , & \text{if } c > (1 - \overline{p}_j)\overline{c}_i \, . \end{cases}$$

As a result, when c is sufficiently high, we have an equilibrium in which

$$\overline{p}_i = \frac{(1-\mu)\Delta_i c - (1-\mu)^2 \Delta_i \Delta_j}{c^2 - (1-\mu)^2 \Delta_i \Delta_j} \, .$$

We first note that the interior equilibrium requires $c > (1-\mu)\Delta_1 \geq (1-\mu)\Delta_2$, as otherwise we cannot get the equilibrium in which both $\overline{p}_1$ and $\overline{p}_2$ are strictly positive. We also get $\frac{\partial \overline{p}_i}{\partial \Delta_i} > 0$ and $\frac{\partial \overline{p}_i}{\partial \Delta_j} < 0$. The exact condition of sufficiently high cost arises from satisfying

$c > (1 - \overline{p}_j)\overline{c}_i$. Using explicit expressions for $\overline{p}_1$ and $\overline{p}_2$, we can show that these conditions are equivalent to requiring

$$c > \frac{\overline{c}_1 + \sqrt{\overline{c}_1^2 - \overline{c}_1\overline{c}_2\kappa(\mu)}}{2} := \overline{c} \geq (1-\mu)\Delta_1 \, ,$$

where $\kappa(\cdot)$ is defined above. As a result, we get the interior equilibrium above if and only if $c > \overline{c}$. Otherwise, the only equilibria are in which one of $\overline{p}_1$, $\overline{p}_2$ is given by Proposition 3, while the other is zero.

Note that $\overline{c} < \overline{c}_1$. As a result, when $c \in (\overline{c}, \overline{c}_1]$, the joint maximum verification intensity of the interior equilibrium is strictly below the intensity of one achieved by the sole fact-checker 1. Now we consider $c > \overline{c}_1$. Using explicit expressions for $\overline{p}_1$ and $\overline{p}_2$, we get

$$\overline{p} = 1 - (1-\overline{p}_1)(1-\overline{p}_2) = 1 - \frac{c^2(c-(1-\mu)\Delta_1)(c-(1-\mu)\Delta_2)}{(c^2-(1-\mu)^2\Delta_1\Delta_2)^2} \, .$$

As a result, our calculations and conclusions coincide for SUE and SFE, with $1-\mu$ in SFE in place of ω in SUE. This completes the proof for SFE.